Tewari's space-vortex electron

The electron as a vortex of a space fluid around a void, compared with measurement — register entry H13

kronspice simulation report · 2026-11-07 · Gaussian units; measured values CODATA 2018 and NIST ASD

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Summary

P. Tewari's 2018 paper (*Physics Essays* 31, 108) describes the electron as an irrotational vortex of an incompressible, massless “space fluid” whose speed reaches the speed of light c at the surface of a central spherical void of radius r_e. From that void it states the electron's charge, mass, angular momentum and magnetic moment, and an electron orbit for hydrogen. This example puts those statements on the electron test bench of *examples/kelvin_ring* and compares each with measurement, with the Dirac electron and QED as the standard counterpart beside every row.

- **The g-factor is a real, parameter-free prediction:** g = 15/8 = 1.875 from eqs. 15–16 for any void radius, against the measured 2.00232, 6.4 % off a value known to about 10^{-13}. With the standard ring moment ½qvρ it is 15/16 = 0.9375.
- **The proton charge** comes out as 12 e (eq. 30); atoms are neutral to ~10^{-21} e.
- **The fine-structure constant** comes out as 15π²/16 = 9.253, 1268 times the measured 7.297×10^{-3}.
- **A density limit** of 3420 g/cm³ that white dwarfs (Sirius B, ~2×10^{6} g/cm³) exceed.
- **The hydrogen orbit gives the wrong light:** speed 0.69 c and orbit frequency 3.405×10^{18} Hz (X-rays), 1381 times the Lyman-α frequency.
- **Status: untested.** The paper gives no Zeeman prediction, the bench's discriminating test.

The hypothesis and the standard prediction

Hypothesis (H13). The electron is a vortex of an incompressible space fluid around a void (Tewari, Physics Essays 31, 108, 2018). His reactionless generator, the other half of H13, is analysed in *examples/space_power_generator*.

Standard prediction. The Dirac electron with QED corrections, which gives the measured values in every row below: g = 2.00232, spin angular momentum ħ/2, and hydrogen with ground-state speed αc and ultraviolet Lyman lines.

The construction

The paper's equations, in its own units (charge in cm³/s, mass in cm⁴/s):

```
q_e = pi^2 r_e^2 c          (eq. 4)
m_e = (4 pi/3) r_e^3 c      (eq. 6)
L   = (4/5) m_e c r_e       (eq. 15, angular momentum)
mu  = (3/4) q_e c r_e       (eq. 16, magnetic moment)
```

- r_e is fixed from the measured charge by eq. 4, as in the paper: r_e = 4.029×10^{-11} cm.
- The paper defines its gram from the measured electron mass (eqs. 10–12), so charge and mass are both inputs.
- Eq. 16 as printed has the dimensions of an SI moment; the Gaussian moment of the same loop is that divided by c, and the g-factor is the same in either reading.
- Measured values: CODATA 2018 and the NIST Atomic Spectra Database (Gaussian units).

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Results

quantity	model	measured	ratio	note
charge (eq. 4)	4.803×10^{-10} statC	4.803×10^{-10}	1	input: r_e fitted to e
void radius vs classical radius	4.029×10^{-11} cm	2.818×10^{-13}	143	no measured size; scattering bounds below about 10^{-18} cm
angular momentum (eq. 15) vs ħ/2	8.802×10^{-28} erg·s	5.273×10^{-28}	1.669	
magnetic moment (eq. 16)	1.451×10^{-20} erg/G	9.285×10^{-21}	1.563	$\lvert\mu_e\rvert = (g/2)\,\mu_B$
g-factor (eqs. 15–16)	**1.875 = 15/8**	**2.00232**	**0.9364**	**independent of r_e**
g with the ring moment ½qvρ	0.9375 = 15/16	2.00232	0.4682	p. 115 integrates dq·v·ρ; a ring current's moment is half that
proton charge (eq. 30)	12 e	e	12	atoms neutral to ~10^{-21} e
fine-structure constant (UPSM eq. 5.2)	$15\pi^2/16 = 9.253$	7.297×10^{-3}	1268	book p. 130 blames experiment; α is measured to 1.5×10^{-10}
density limit (UPSM §2.8) vs Sirius B	3420 g/cm³	~2×10^{6}	0.0017	white dwarfs exceed it
hydrogen orbit radius (eq. 35) vs a_0	9.67×10^{-10} cm	5.292×10^{-9}	0.183	
hydrogen orbit speed vs αc	2.069×10^{10} cm/s (0.69 c)	2.188×10^{8}	94.6	
orbit frequency vs Lyman-α	3.405×10^{18} Hz	2.466×10^{15}	1381	a classical orbit radiates at v/2πr

Ratio = model / measured. Measured values in the same units as the model column. a_0: Bohr radius; α: fine-structure constant; μ_B: Bohr magneton.

- **The g-factor is a real prediction.** $g = 2m_e c\mu/(eL)$ from eqs. 15–16, and r_e, q and m all cancel, so 15/8 (or 15/16 with the ring moment) holds for any void radius. It is 6.4 % from the measured value, which is known to about 10^{-13}.
- **The hydrogen orbit gives the wrong light.** A charge orbiting at 0.69 c on 24 r_e radiates X-rays; hydrogen's lines are ultraviolet and its ground state has v = αc.
- **Zeeman effect: not treated in the paper.** The bench's discriminating test needs the model's prediction for the splitting and polarisation of a line in a field B (measured: $\Delta\omega = eB/2m_e c$ per unit of m_l, σ± circularly polarised).

Validation

Before recording the comparisons, the test reproduces the paper's own numbers where it states them: $r_e \approx 4\times10^{-11}$ cm, eq. 4 giving 4.7326×10^{-10} at that radius, 1 g = 8.8×10^{6} cm^4/s by eq. 11, and a void volume of 2.67×10^{-31} cm³. *main_test.go* also checks g = 15/8 and $\alpha = 15\pi^2/16$.

Limits of the analysis

- The bench takes the paper's equations as printed; r_e is fitted to the measured charge and the mass enters through the paper's own definition of the gram, so charge and mass are inputs, not predictions.

Status

Register status: **untested**, pending a Zeeman prediction from the model. The register records g = 15/8 (15/16 with the standard ring moment) against 2.0023, proton charge 12e, $\alpha = 15\pi^2/16$ against 1/137, and a density limit white dwarfs exceed, as the values the model must also meet.

Reproducing the results

```
go run ./examples/space_vortex
go test ./examples/space_vortex
```

Sources: P. Tewari, Physics Essays 31(1), 108–129 (2018), doi:10.4006/0836-1398-31.1.108; P. Tewari, *Universal Principles of Space and Matter* (2017), eq. 5.2 and §2.8; CODATA 2018 (E. Tiesinga et al., Rev. Mod. Phys. 93, 025010, 2021); NIST Atomic Spectra Database. Notes and page checks in doc/tewari-references.md.

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